Palitizer / PLC_1 [CPU 1211C DC/DC/DC] / Program blocks

FB3_Elevator_Stacker [FB3]

FB3_Elevator_Stacker Properties							
General							
Name	FB3_Elevator_Stacker	**Number**	3	**Type**	FB	**Language**	SCL
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

FB3_Elevator_Stacker									
Name	**Data type**	**Default value**	**Retain**	**Accessible from HMI/OPC UA/Web API**	**Writable from HMI/OPC UA/Web API**	**Visible in HMI engineering**	**Setpoint**	**Supervision**	**Comment**
▼ Input									
▼ i_stState	"UDT_SysStatus"		Non-retain	True	True	True	False		
xModeAuto	Bool	false	Non-retain	True	True	True	False		
xModeManual	Bool	false	Non-retain	True	True	True	False		
xRunning	Bool	false	Non-retain	True	True	True	False		
xPaused	Bool	false	Non-retain	True	True	True	False		
xFaulted	Bool	false	Non-retain	True	True	True	False		
iStateID	Int	0	Non-retain	True	True	True	False		
iCurrentLayer	Int	0	Non-retain	True	True	True	False		
iTargetLayers	Int	0	Non-retain	True	True	True	False		
iTotalBoxes	DInt	0	Non-retain	True	True	True	False		
iTotalPallets	DInt	0	Non-retain	True	True	True	False		
wAlarmWord	Word	16#0	Non-retain	True	True	True	False		
▼ i_stCmd	"UDT_SysCommand"		Non-retain	True	True	True	False		
xSwAuto	Bool	false	Non-retain	True	True	True	False		
xSwManual	Bool	false	Non-retain	True	True	True	False		
xBtnStart	Bool	false	Non-retain	True	True	True	False		
xBtnStop	Bool	false	Non-retain	True	True	True	False		
xBtnReset	Bool	false	Non-retain	True	True	True	False		
xEStop_OK	Bool	false	Non-retain	True	True	True	False		
xHmiResetCounters	Bool	false	Non-retain	True	True	True	False		
xHmiAckFault	Bool	false	Non-retain	True	True	True	False		
Output									
▼ InOut									
▼ io_stState	"UDT_SysStatus"			False	False	False	False		
xModeAuto	Bool			False	False	False	False		
xModeManual	Bool			False	False	False	False		
xRunning	Bool			False	False	False	False		
xPaused	Bool			False	False	False	False		
xFaulted	Bool			False	False	False	False		
iStateID	Int			False	False	False	False		
iCurrentLayer	Int			False	False	False	False		
iTargetLayers	Int			False	False	False	False		
iTotalBoxes	DInt			False	False	False	False		
iTotalPallets	DInt			False	False	False	False		
wAlarmWord	Word			False	False	False	False		
▼ Static									
stat_Step	Int	0	Non-retain	True	True	True	False		
▼ stat_Retract_Timer	IEC_TIMER		Non-retain	True	True	True	False		
PT	Time	T#0ms	Non-retain	True	True	True	False		
ET	Time	T#0ms	Non-retain	True	False	True	False		
IN	Bool	false	Non-retain	True	True	True	False		
Q	Bool	false	Non-retain	True	False	True	False		
stat_Row_Count	Int	0	Non-retain	True	True	True	True		
stat_horizental_box_count	Int	0	Non-retain	True	True	True	False		
stat_vertical_box_count	Int	0	Non-retain	True	True	True	False		
stat_Box_Count	Int	0	Non-retain	True	True	True	False		
stat_Layer_Count	Int	0	Non-retain	True	True	True	False		
stat_Target_Boxes	Int	0	Non-retain	True	True	True	False		
stat_Target_Rows	Int	0	Non-retain	True	True	True	False		
stat_Box_Sensor_Prev	Bool	false	Non-retain	True	True	True	False		
stat_Fault_ID	Int	0	Non-retain	True	True	True	False		
▼ stat_Push_Timeout	IEC_TIMER		Non-retain	True	True	True	False		
PT	Time	T#0ms	Non-retain	True	True	True	False		
ET	Time	T#0ms	Non-retain	True	False	True	False		
IN	Bool	false	Non-retain	True	True	True	False		
Q	Bool	false	Non-retain	True	False	True	False		

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/OPC UA/Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ stat_Plate_Timeout	IEC_TIMER		Non-retain	True	True	True	True		
PT	Time	T#0ms	Non-retain	True	True	True	False		
ET	Time	T#0ms	Non-retain	True	False	True	False		
IN	Bool	false	Non-retain	True	True	True	False		
Q	Bool	false	Non-retain	True	False	True	False		
▼ stat_Eject_Timeout	IEC_TIMER		Non-retain	True	True	True	True		
PT	Time	T#0ms	Non-retain	True	True	True	False		
ET	Time	T#0ms	Non-retain	True	False	True	False		
IN	Bool	false	Non-retain	True	True	True	False		
Q	Bool	false	Non-retain	True	False	True	False		
▼ push	TON_TIME		Non-retain	True	True	True	True		
PT	Time	T#0ms	Non-retain	True	True	True	False		
ET	Time	T#0ms	Non-retain	True	False	True	False		
IN	Bool	false	Non-retain	True	True	True	False		
Q	Bool	false	Non-retain	True	False	True	False		
▼ EjectTimeout	TON_TIME		Non-retain	True	True	True	True		
PT	Time	T#0ms	Non-retain	True	True	True	False		
ET	Time	T#0ms	Non-retain	True	False	True	False		
IN	Bool	false	Non-retain	True	True	True	False		
Q	Bool	false	Non-retain	True	False	True	False		
Temp									
Constant									

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// --- 1. Fault Reset Logic ---
// Operator must clear the jam and press Reset to clear the Fault ID
IF ("I_Reset" OR #i_stCmd.xHmiAckFault)  AND "I_E_Stop" THEN
  #stat_Fault_ID := 0;
  "M_Machine_Fault" := FALSE;
END_IF;
// --- 2. Diagnostics & Watchdog Timers ---
// E-Stop Monitor
IF NOT "I_E_Stop" THEN
  #stat_Fault_ID := 1;
  #io_stState.xFaulted := TRUE;
  #stat_Step := 0;
END_IF;
// Watchdog Timers (Running parallel to the state machine)
#push(IN:=#stat_Step = 10,
      PT:=T#2.0S);

#EjectTimeout(IN := #stat_Step = 81,
              PT := T#8.0S);


#stat_Plate_Timeout.IN := #stat_Step = 60;
#stat_Plate_Timeout.PT := T#4.0S;

// Fault Triggers (If a timer finishes, assign the specific Fault ID)
IF #push.Q THEN
  #stat_Fault_ID := 2;
  #io_stState.xFaulted := TRUE;
  #stat_Step := 0;
END_IF;
IF #stat_Plate_Timeout.Q THEN
  #stat_Fault_ID := 3;
  #io_stState.xFaulted := TRUE;
  #stat_Step := 0;
END_IF;
IF #EjectTimeout.Q THEN
  #stat_Fault_ID := 4;
  #io_stState.xFaulted := TRUE;
  #stat_Step := 0;
END_IF;
// --- Alarm Word Mapping (Slice Access) ---
// If the Fault ID equals 1, turn on Bit 1 of the Word.
#io_stState.wAlarmWord.%X1 := (#stat_Fault_ID = 1);
#io_stState.wAlarmWord.%X2 := (#stat_Fault_ID = 2);
#io_stState.wAlarmWord.%X3 := (#stat_Fault_ID = 3);
#io_stState.wAlarmWord.%X4 := (#stat_Fault_ID = 4);
// --- Safety & Mode Interlock ---
IF NOT #i_stState.xRunning THEN

  "Cmd_Push_Auto" := FALSE;
  "Cmd_Plate_Auto" := FALSE;
  "Cmd_ElevUp_Auto" := FALSE;
  "Cmd_ElevDown_Auto" := FALSE;
  "Q_Move_to_Limit" := FALSE;
  "M_Reset_Row" := FALSE;
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  "M_Eject_Pallet" := FALSE;
  "Cmd_Clamp_Auto" := FALSE;
  #stat_Row_Count := 0;
  #stat_Layer_Count := 0;
  RETURN;
END_IF;
// --- Maintenance Reset ---
IF #i_stCmd.xHmiResetCounters THEN
  #stat_Layer_Count := 0;
  #stat_Row_Count := 0;
  #io_stState.iTotalBoxes := 0;
  #io_stState.iTotalPallets := 0;
  #stat_Step := 0; // Safely drops the machine back to the IDLE state
END_IF;

// --- Timer Execution ---
"IEC_Timer_0_DB_1".TON(IN := (#stat_Step = 20),
             PT := T#1.5S); // Retract
"IEC_Timer_0_DB_2".TON(IN := (#stat_Step = 10),
             PT := T#0.5S); // Push
"plate_open".TON(IN := (#stat_Step = 40),
         PT := T#3.0S);
"plate_close".TON(IN := (#stat_Step = 60),
          PT := T#2.0S); // FIXED: Changed to Step 60
"Elevator_Timer".TON(IN := (#stat_Step = 52),
           PT := T#500MS);
// --- Dynamic Pattern Generator ---
// Even Layers (0, 2): 3 boxes per row, 2 rows. TURN OFF.
IF (#stat_Layer_Count MOD 2) = 0 THEN
  "M_Target_Boxes" := 2;
  "M_Target_Rows" := 3;
  "M_Enable_Turn" := FALSE;
  // Odd Layers (1, 3): 2 boxes per row, 3 rows. TURN ON.
ELSE
  "M_Target_Boxes" := 3;
  "M_Target_Rows" := 2;
  "M_Enable_Turn" := TRUE;
END_IF;

// --- The Core State Machine ---
CASE #stat_Step OF

  0:  // IDLE STATE
    "Cmd_Push_Auto" := FALSE;
    "M_Reset_Row" := FALSE;
    "Cmd_Plate_Auto" := FALSE;
    "Cmd_ElevUp_Auto" := FALSE;
    "Cmd_ElevDown_Auto" := FALSE;
    "Q_Move_to_Limit" := FALSE;
    "M_Eject_Pallet" := FALSE;

    // Force the elevator to absolute top for a fresh pallet
    IF #stat_Layer_Count = 0 AND "I_Pallet_Loaded" THEN
      "Q_Move_to_Limit" := TRUE;
      "Cmd_ElevUp_Auto" := TRUE;
    END_IF;

    // Proceed ONLY if row is ready, pallet is present, and elevator is at the top
    IF "M_Row_Ready" AND "I_Pallet_Loaded" AND NOT "I_Elevator_Moving" THEN
      "Q_Move_to_Limit" := FALSE;
      "Cmd_ElevUp_Auto" := FALSE;
      // FIXED: Removed premature Elevator Down command here
      #stat_Step := 10;
    END_IF;

  10: // PUSH STATE
    "Q_Box_Feeder" := FALSE;
    "Cmd_Push_Auto" := TRUE;

    IF "IEC_Timer_0_DB_2".Q THEN
      #stat_Step := 20;
    END_IF;

  20: // RETRACT STATE
    "Q_Box_Feeder" := FALSE;
    "Cmd_Push_Auto" := FALSE;
    "M_Reset_Row" := TRUE;

    IF "IEC_Timer_0_DB_1".Q THEN
      #stat_Step := 30;
    END_IF;

  30: // ROW MATH (FIXED: Single Scan Execution)
    "Q_Box_Feeder" := TRUE;
    "M_Reset_Row" := FALSE;
    #stat_Row_Count := #stat_Row_Count + 1; // Increment happens exactly once
    // Add boxes to the total
    #io_stState.iTotalBoxes := #io_stState.iTotalBoxes + "M_Target_Boxes";
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    IF #stat_Row_Count >= "M_Target_Rows" THEN
      #stat_Layer_Count := #stat_Layer_Count + 1;
      #stat_Step := 31; // Jump to new state to wait for clamp
    ELSE
      #stat_Step := 0;  // Go back for next row
    END_IF;

  31: // WAIT FOR CLAMP (Prevents Math Loop Bug)
    "Cmd_Clamp_Auto" := TRUE;

    IF "I_Clamped" THEN
      #stat_Step := 40;
    END_IF;

  40: // DROP LAYER (OPEN PLATE)
    "Cmd_Clamp_Auto" := TRUE;

    "Cmd_Plate_Auto" := TRUE;

    IF "plate_open".Q THEN

      #stat_Step := 50;
    END_IF;

  50: // INDEX 1 START (FIXED: Race condition protection restored)
    "Cmd_ElevUp_Auto" := FALSE;
    "Q_Move_to_Limit" := FALSE;
    "Cmd_ElevDown_Auto" := TRUE;

    IF "I_Elevator_Moving" THEN // Wait for motor to physically start
      #stat_Step := 51;
    END_IF;

  51: // INDEX 1 FINISH
    IF NOT "I_Elevator_Moving" THEN
      "Cmd_ElevDown_Auto" := FALSE;
      #stat_Step := 52;
    END_IF;

  52: // PAUSE
    IF "Elevator_Timer".Q THEN
      #stat_Step := 54;
    END_IF;

  53: // INDEX 2 START
    "Cmd_ElevDown_Auto" := TRUE;
    IF "I_Elevator_Moving" THEN
      #stat_Step := 54;
    END_IF;

  54: // INDEX 2 FINISH
    IF NOT "I_Elevator_Moving" THEN
      "Cmd_ElevDown_Auto" := FALSE;
      #stat_Step := 60;
    END_IF;

  60: // CLOSE PLATE
    "Cmd_Plate_Auto" := FALSE;
    IF "plate_close".Q THEN
      "Cmd_Clamp_Auto" := FALSE;
      #stat_Row_Count := 0;
      // "M_Reset_Row" := TRUE;
      #stat_Step := 70;
    END_IF;

  70: // LAYER DECISION
    IF #stat_Layer_Count >= 4 THEN
      #stat_Step := 80;
    ELSE
      #stat_Step := 0;
    END_IF;

  80: // DISCHARGE PALLET (FIXED: Race condition protection restored)
    "Q_Move_to_Limit" := TRUE;
    "Cmd_ElevDown_Auto" := TRUE;

    IF "I_Elevator_Moving" THEN // Wait for motor to start
      #stat_Step := 81;
    END_IF;

  81: // WAIT FOR DISCHARGE
    IF NOT "I_Elevator_Moving" THEN
      "M_Eject_Pallet" := TRUE;
      "Cmd_ElevDown_Auto" := FALSE;
      "Q_Move_to_Limit" := FALSE;

      IF NOT "I_Pallet_Loaded" THEN
        #stat_Layer_Count := 0;
```

```
        "M_Eject_Pallet" := FALSE;
        #stat_Step := 0;
        #io_stState.iTotalPallets := #io_stState.iTotalPallets + 1;
      END_IF;
    END_IF;

END_CASE;

// --- HMI Data Reporting ---
#io_stState.iCurrentLayer := #stat_Layer_Count;
#io_stState.iTargetLayers := 4; // Since logic ejects at layer 4
```

Symbol	Address	Type	Comment
"Cmd_Clamp_Auto"	%M3.0	Bool	
"Cmd_ElevDown_Auto"	%M4.4	Bool	
"Cmd_ElevUp_Auto"	%M4.2	Bool	
"Cmd_Plate_Auto"	%M3.2	Bool	
"Cmd_Push_Auto"	%M0.6	Bool	
"Elevator_Timer".Q		Bool	
"I_Clamped"	%I0.7	Bool	Sensor confirming the clamp is fully closed.
"I_E_Stop"	%I0.4	Bool	
"I_Elevator_Moving"	%I0.5	Bool	
"I_Pallet_Loaded"	%I1.3	Bool	Sensor ON the elevator. Goes TRUE when pallet is in position.
"I_Reset"	%I1.5	Bool	
"IEC_Timer_0_DB_1".Q		Bool	
"IEC_Timer_0_DB_2".Q		Bool	
"M_Eject_Pallet"	%M0.4	Bool	
"M_Enable_Turn"	%M1.0	Bool	Flag to activate the Turn actuator.
"M_Machine_Fault"	%M1.1	Bool	
"M_Reset_Row"	%M0.3	Bool	
"M_Row_Ready"	%M0.2	Bool	
"M_Target_Boxes"	%MW10	Int	Tells the counter how many boxes to wait for.
"M_Target_Rows"	%MW12	Int	Tells the sequencer how many rows to push.
"plate_close".Q		Bool	
"plate_open".Q		Bool	
"Q_Box_Feeder"	%Q0.4	Bool	First conveyor bringing boxes in.
"Q_Move_to_Limit"	%Q1.4	Bool	
#EjectTimeout		IEC_Timer	
#EjectTimeout.Q		Bool	
#i_stCmd.xHmiAckFault		Bool	
#i_stCmd.xHmiResetCounters		Bool	
#i_stState.xRunning		Bool	
#io_stState.iCurrentLayer		Int	
#io_stState.iTargetLayers		Int	
#io_stState.iTotalBoxes		DInt	
#io_stState.iTotalPallets		DInt	
#io_stState.wAlarmWord.%X1		Bool	
#io_stState.wAlarmWord.%X2		Bool	
#io_stState.wAlarmWord.%X3		Bool	
#io_stState.wAlarmWord.%X4		Bool	
#io_stState.xFaulted		Bool	
#push		IEC_Timer	
#push.Q		Bool	
#stat_Fault_ID		Int	
#stat_Layer_Count		Int	
#stat_Plate_Timeout.IN		Bool	
#stat_Plate_Timeout.PT		Time	
#stat_Plate_Timeout.Q		Bool	
#stat_Row_Count		Int	
#stat_Step		Int	